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Course: Networks System Design

Instructor: KUY Movsun

Assignment: Lab10 (Network Layer Control Plane Algorithms)

Due Date: January, 13 2026, (12:00 AM)

Part1: The Link State Walkthrough (Dijkstra)

Step	N' (Nodes added)	D(B), p(B)	D(C), p(C)	D(D), p(D)	D(E), p(E)	D(F), p(F)
0	A	2, A	1, A	∞	∞	∞
1	A, C	2, A	1, A	3, C	5, C	∞
2	A, C, B	2, A	1, A	3, C	5, C	∞
3	A, C, B, D	2, A	1, A	3, C	4, D	8, D
4	A, C, B, D, E	2, A	1, A	3, C	4, D	5, E
5	A, C, B, D, E, F	2, A	1, A	3, C	4, D	5, E

Final answer

```
Path to F: A → C → D → E → F

Total cost: 5
```

Part 2: Count-to-Infinity Simulation

Initial State

Router	Cost to Z	Next Hop
Y	1	Direct
X	5	Y

Iteration 1

Router	Update Action	New Cost to Z	Next Hop
Y	Detects link change (60). Chooses route via X: 4 + 5 = 9	9	X
X	Receives Y's update (9). Updates: 4 + 9 = 13	13	Y

Iteration 2

Router	Update Action	New Cost to Z	Next Hop
Y	Receives X's update (13). Updates: 4 + 13 = 17	17	X
X	Receives Y's update (17). Updates: 4 + 17 = 21	21	Y

Explanation

- Costs keep increasing step by step toward infinity.
- This slow convergence is why the phenomenon is called **"Bad news travels slow"** in Distance Vector routing.

Part 3: Packet Tracer Activity (RIP)

Objective

Build a 3-hop network and verify how RIP (a Distance Vector protocol) handles hop counts.

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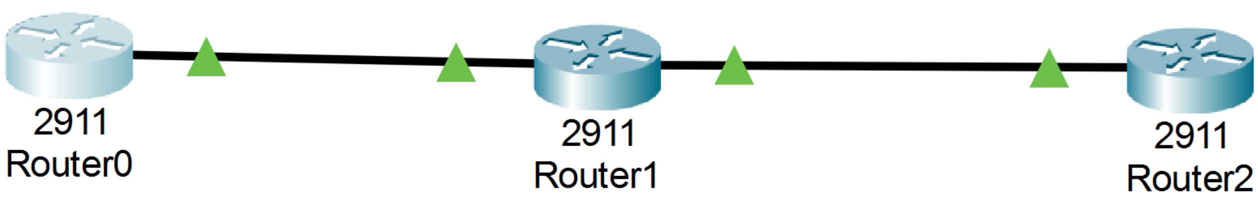
1. Setup Topology

Logical map:

- Devices: Cisco 2911 Routers
- Connections: GigabitEthernet cables



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2. IP Configuration

Router	Interface	IP Address	Subnet Mask
R0	G0/0	192.168.1.1	255.255.255.0

Router	Interface	IP Address	Subnet Mask
R1	G0/0	192.168.1.2	255.255.255.0
R1	G0/1	192.168.2.1	255.255.255.0
R2	G0/0	192.168.2.2	255.255.255.0
R2	Loopback0	10.10.10.1	255.255.255.0

Loopback creation:

```
Router(config)# interface loopback 0
Router(config-if)# ip address 10.10.10.1 255.255.255.0
```

3. RIP Configuration

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# no auto-summary
Router(config-router)# network 192.168.1.0
Router(config-router)# network 192.168.2.0
Router(config-router)# network 10.0.0.0 (only on Router2)
```

4. Verification

1. Run show ip route. Look for the line starting with R (RIP) for destination 10.0.0.0/8.

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 1 subnets
R       10.10.10.0/24 [120/2] via 192.168.1.2, 00:00:02, GigabitEthernet0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
R    192.168.2.0/24 [120/1] via 192.168.1.2, 00:00:02, GigabitEthernet0/0

Router#
```

2. What does the 120 represent? What does the 2 represent?

- **120** is the **Administrative Distance (AD)** of RIP.
  - It defines the trust level of the routing source.
  - RIP has a default AD of 120, which is relatively low priority compared to other protocols like OSPF (110) or EIGRP (90).
- **2** is the **hop count** to reach the destination network.
  - It means there are **2 routers** between the current router and the destination `10.10.10.0/24`.
  - This matches the topology: Router0 → Router1 → Router2.

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3. How many lines of output do you see in traceroute? Does this match the metric?

```
Router#traceroute 10.10.10.1
Type escape sequence to abort.
Tracing the route to 10.10.10.1

  1   192.168.1.2     0 msec    0 msec    0 msec
  2   192.168.2.2     0 msec    0 msec    0 msec
Router#
```

- The `traceroute 10.10.10.1` command shows **2 lines of output**:

1. First hop: `192.168.1.2` (Router1)
2. Second hop: `192.168.2.2` (Router2)

- Yes, this matches the RIP metric of **2 hops** shown in the routing table `[120/2]`.